

Time-Domain Modeling of an RF All-Digital PLL

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Abstract—A new phase-domain all-digital phase-locked loop (ADPLL) for RF wireless applications has recently been proposed and commercially demonstrated. In this brief, we propose time-domain modeling and simulation techniques of the ADPLL that are well suited for system analysis using high-level programming languages, e.g., Matlab. They are based on the event-driven principles inherent in hardware description languages, e.g., VHDL, and enable the development of accurate and time-efficient behavioral models. The proposed techniques are demonstrated and validated through experimental results for a GSM standard.

Index Terms—All-digital phase-locked loop (ADPLL), CMOS, digitally controlled-oscillator (DCO), event driven, GSM, mobile phones, phase detection, phase noise, simulation, time-domain modeling, time-to-digital-converter (TDC), wireless.

I. INTRODUCTION

REDUCTIONS of production cost and power consumption are two main objectives of today's active research related to the cellular phone industry. Advanced low-voltage nanometer-scale (< 100 nm feature size) CMOS processes allow for a large degree of scalability and integration of digital circuitry but greatly increase the design cycle of pure analog/RF components, e.g., used in charge-pump phase-locked loop (PLL) frequency synthesizers, due to a poor process characterization and parameter spread.

By exploring the fine time-domain resolution of a nanometer-scale CMOS process, a phase-domain all-digital PLL (ADPLL)-based RF frequency synthesizer and transmitter, having a digitally controlled oscillator (DCO) and a time-to-digital converter (TDC) as core components, were recently proposed and commercially demonstrated [1], [2]. A simplified block diagram of the phase-domain ADPLL architecture is shown in Fig. 1. The system accepts as input a stable frequency reference (FREF) clock signal of frequency f_R (typically 8–40 MHz) and generates a variable clock (CKV) signal of frequency f_V (typically in the multigigahertz range) as defined by the frequency command word (FCW). FCW is a real number and is defined as the expected number of CKV clock cycles that fit into one FREF clock cycle, i.e., $FCW \triangleq E\{f_V\}/f_R$ [3]. The phase detector is a simple adder that operates on signals that carry phase information. Due to the existence of the two asynchronous (in general) clock domains, FREF and CKV, retiming is performed by oversampling the FREF clock with CKV for synchronization

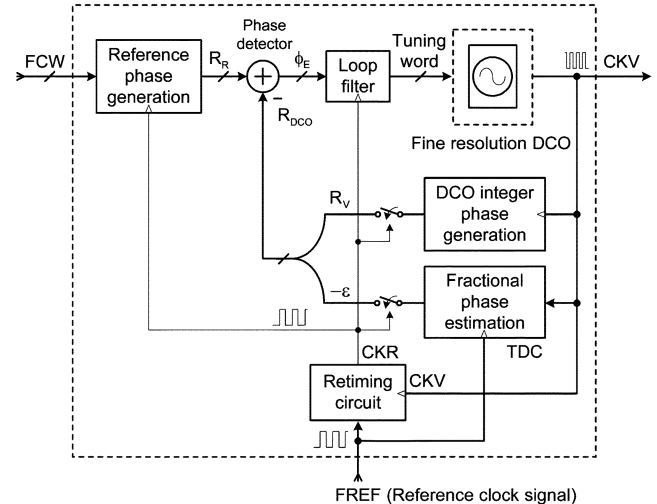


Fig. 1. Block diagram of the phase-domain ADPLL frequency synthesizer.

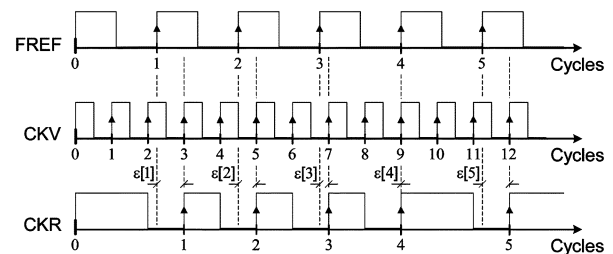


Fig. 2. Retiming of the reference clock signal (FREF) rising transition to the next rising transition of the DCO output clock signal (CKV) ($FCW = 2.25$).

purposes (see Fig. 2) [3]. The retimed clock, CKR, of average frequency f_R is used to synchronize the internal ADPLL operations. However, the retiming process generates a fractional phase error ϵ (in CKV cycles). This error is estimated using a TDC [4], and it needs to be considered in the phase detection mechanism. An overview of the ADPLL phase detection mechanism is given in Section II.

The DCO is an LC -tank-based oscillator having its inductor largely fixed and its capacitor divided into a set of binary switchable MOS varactor banks [5]. Despite the analog-intensive nature of the DCO, event-driven-based behavioral modeling techniques of its phase noise and its period (frequency) perturbation have been reported in [6]. They form the foundation of this work that allows the ADPLL to be modeled, simulated, and validated using event-driven concepts. The event-driven model realization of the DCO, described in [6], shows that the timing information associated with its output clock transitions (events) suffices for its phase noise performance evaluation. On the contrary, SPICE and SpectreRF transistor-level-based simulations of the DCO would generate nonuniform highly oversampled

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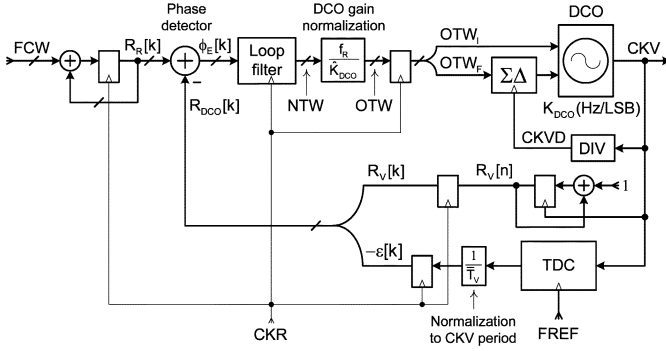


Fig. 3. Details of the phase-domain ADPLL frequency synthesizer.

signals, resulting in an excessive amount of data since longer simulation times are now required for similar phase-noise performance observations.

Main advantage of event-driven simulators is that only the time instances of interest (e.g., edges or transitions) of the underlying clock signals are exercised to control and synchronize the functionality of the simulated system. Furthermore, access to *time* is inherent in such simulators by means of adding delays to signal transitions, activating processes when a signal transition occurs or by means of *time*-type data objects, as in VHDL. The direction of how to develop such *time*-type features that would allow the behavioral modeling and fast simulation of the ADPLL system using high-level programming languages, such as Matlab or C, is a main contribution of this brief.

In this brief, a novel high-level time-domain modeling and simulation technique of the recently proposed phase-domain ADPLL frequency synthesizer is presented and is based on the *event-driven* principles mentioned above. This technique combined with the appropriate minimal run-time complexity behavioral model for each subblock of the ADPLL (e.g., digital loop filter) can be realized using any high-level programming language such as Matlab, C, or SystemC and can rapidly and accurately predict the transient, steady-state, and phase-noise performance of the frequency synthesizer when used in the system-level or integration-level design stage.

II. OVERVIEW OF THE ADPLL OPERATION

Here, we provide a concise overview of the ADPLL principles of operation that will help the reader to easily follow the modeling and simulation techniques proposed in Sections III–VI. A detailed diagram of the ADPLL architecture is shown in Fig. 3 together with the signals reflected in the pseudocode that models the ADPLL behavior (see Fig. 8).

The DCO contains four capacitor banks that are used with three operational modes: process, voltage, and temperature (PVT) variations, acquisition, and tracking. The first two are used during frequency settling and are frozen afterwards [5]. The frequency f_V of the DCO output clock (CKV) signal is controlled by the oscillator tuning word (OTW), which is a fixed-point digital word, i.e., $OTW = OTW_I + OTW_F$, with the fractional part utilized in tracking mode only. The OTW-to- f_V dependency is in general nonlinear because of the LC-tank-based structure of the DCO. The slope of the OTW-to- f_V characteristic represents the DCO gain, K_{DCO} (in Hz/LSB), and its step size determines the DCO frequency

resolution. It indicates a frequency deviation of K_{DCO} (Hz) from the DCO effective center frequency f_0 , when the OTW is changed by 1 LSB. The finest frequency resolution is utilized when the ADPLL is in tracking mode. Specifically, for 90-nm CMOS and small frequency perturbations around 1.8 GHz, the K_{DCO} is about 12 kHz/LSB. However, further resolution improvement is achieved by exercising the fractional part of OTW to dither the tracking bank varactors through high-speed $\Sigma\Delta$ dithering, as shown in Fig. 3. The $\Sigma\Delta$ modulator is implemented as a second-order multistage noise-shaping (MASH)-type structure and is described in [5].

PVT variations as well as the DCO variable oscillation frequency, are a set of factors that highly affect K_{DCO} in an uncertain way. To minimize the ADPLL dependency on the DCO gain variation, K_{DCO} is algorithmically estimated as $\hat{K}_{DCO}$, and the DCO gain is normalized to the stable reference frequency f_R , as shown in Fig. 3 (DCO gain normalization block— $f_R/\hat{K}_{DCO}$).

The ADPLL phase-detection mechanism is based on cycles (i.e., phase) accumulation performed at the rising edges of the CKR and CKV clocks [3]. Two phase accumulators are utilized to generate the reference and the DCO-integer phases, as: $R_R[k] = \sum_{l=1}^k FCW$ and $R_V[n] = \sum_{l=1}^n 1$ (in CKV cycles), respectively, where k and n are the time indexes associated with the rising-edge timestamps of the CKR/FREF and CKV clock signals, respectively. The fractional phase error $\varepsilon[k]$, with $\varepsilon \in (0, 1)$, that is generated due to the retiming process is estimated using a TDC, and it needs to be considered in the phase-detection mechanism. The latter is a simple arithmetic subtraction (linear operation) and is defined as

$$\phi_E[k] \triangleq R_R[k] - R_{DCO}[k] = R_R[k] - (R_V[k] - \varepsilon[k]) \quad (1)$$

where $R_V[k]$ is the DCO-integer phase, sampled at the rising transitions of the CKR clock.

To close the loop, ϕ_E is processed by the digital loop filter, and then it is applied to the normalized DCO by means of the normalized tuning word (NTW) in order to control the phase/frequency of its output clock signal (CKV).

III. ADPLL MODELING AND SIMULATION PRINCIPLES

It is worth mentioning a recent work [7] on behavioral modeling of the proposed phase-domain ADPLL-based frequency synthesizer. The initial objective there is to decouple from the simulation process the DCO output RF rate and the TDC timing resolution Δt_{TES} , as this would require a high sampling frequency to correctly represent the ADPLL signals and therefore a long simulation time. On this basis, a dual-modulus-divider (DMD)-based ADPLL behavioral model was developed, operating at the reference clock (FREF) rate, for the estimation of the timing difference between the FREF and the CKV signals transitions. However, this model provides only a rough estimation of this timing difference, resulting in fractional phase estimations that cause the modeled ADPLL operation, though stable, to significantly deviate from the actual one. Based on the modeling techniques presented in this brief, the exact timing difference between the FREF and the CKV signal transitions is readily available during the simulation process, resulting in accurate behavioral modeling of the ADPLL. Another work worth

mentioning on time-domain modeling of digital PLLs is [8]. It uses Matlab functions to build an event-driven simulation environment, however, at the cost of significant overhead. The techniques presented in this brief derive all of the benefits of the event-driven principle without this penalty.

To model the behavior of the ADPLL, it suffices to consider only the transitions (conventionally rising) of the clock signals pertaining to its synchronization, i.e., FREF, CKV, and CKR, together with their associated rising-edge timestamps (event time instances). Considering each clock domain separately, the time instances of the rising clock edges can be identified by progressively increasing pointers (indexes) indicating the completion of full FREF, CKV, and CKR clock cycles. These indexes can be used to properly synchronize the data flow and model the digital portion of the ADPLL, i.e., the reference phase generation, the DCO integer phase generation, the phase detector, the loop filter, the $\Sigma\Delta$ modulator, as shown in Fig. 3, and any other digital circuitry related to the ADPLL operation and calibration. Having available those indexes, we can track the actual timing evolution of the various clock domains by employing the time-domain modeling techniques. This allows us to create timestamp data for the FREF and the CKV clocks on-the-fly and account for the fact that both clock sources exhibit phase noise, which causes the signals transitions to vary from the ideal ones, usually in an unpredicted way. In this way, we can properly model the TDC and therefore the retiming process since the timing information of both FREF and CKV clock domains is readily available.

Highly accurate, versatile, and quick-to-simulate ADPLL models can be developed by applying the aforementioned event-driven-based modeling and simulation techniques using any high-level programming language. Such models form the basis for a multidimensional study of the ADPLL, related to its transient, steady-state, and phase noise performance.

IV. TIME-DOMAIN VERSUS FREQUENCY-DOMAIN MODELING

A frequency-domain (single-rate z -domain) model of the ADPLL frequency synthesizer that operates on the excess, rather than the absolute, phase values is shown in Fig. 4. It also includes the two primary phase noise sources $\varphi_{n,TDC}$ and $\varphi_{n,DCO}$, which are due to the TDC and DCO phase noise and nonlinearities (e.g., finite resolution, LC -tank quantization), respectively.

The main disadvantage of the z -domain model is that it considers the ADPLL as a single-rate system with uniform sampling intervals ($1/f_R$, in the range of 8–40 MHz), and ignores the coexistence of the CKV clock domain (in the range of 1600–2000 MHz for a GSM system). This poses limitations on the proper modeling and simulation of the RF-rate phase noise of the DCO, which now needs to be expressed with respect to the reference rate with unavoidable frequency aliasing. Furthermore, the TDC-induced phase noise, which highly affects the spurious tone performance of the ADPLL, cannot be accurately modeled since the CKV and FREF clock timestamps are not available for the estimation of the fractional phase error (ε) due to retiming and the associated phase noise due to the limited time resolution of the TDC. However, the z -domain model generates a fractional phase feedback as a

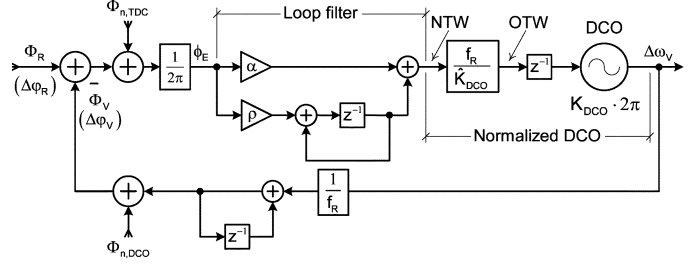


Fig. 4. z -domain ADPLL model: type-II second-order configuration.

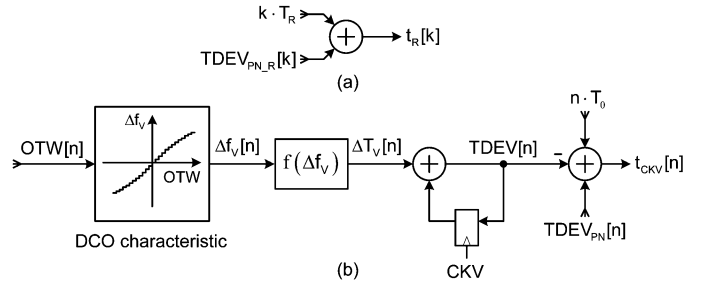


Fig. 5. Time-domain models for the generation of (a) reference (FREF) and (b) DCO-based (CKV) clock timestamps.

result of its closed-loop dynamics that does not necessarily reflect the actual fractional phase error (ε), but the resulting closed-loop behavior is adequate for the prediction of the filtering performance of the ADPLL at low frequency offsets from the carrier.

By starting with the time-domain modeling techniques of an oscillator suggested in [6], we can generate timestamp data for both FREF and CKV clock domains that allow the development of multirate ADPLL models that accurately predict the performance of the actual system. Time-domain models for the FREF and the DCO-based CKV clock domains are shown in Fig. 5. The FREF clock timestamps are generated by creating timing intervals of length equal to the reference clock period $T_R = 1/f_R$ and adding perturbation $TDEV_{PN,R}$ related to the phase noise of the reference clock source. On the other hand, the generation of the CKV clock timestamps needs also to consider the loop control by means of OTW , which tunes the CKV clock frequency f_V according to the OTW -to- f_V characteristic of the DCO. Frequency deviations Δf_V away from f_0 result in time deviations ΔT_V away from the corresponding clock period $T_0 = 1/f_0$ according to

$$\Delta T_V = \frac{\Delta f_V}{f_0 \cdot (f_0 + \Delta f_V)}. \quad (2)$$

Finally, frequency deviations for which the DCO gain can be considered constant, are expressed as $\Delta f_V \simeq K_{DCO} \cdot OTW$.

V. ADPLL MULTIRATE TIME-DOMAIN MODELING

Here, we propose modeling and simulation techniques for the phase-domain ADPLL that can be realized using any high-level programming language such as Matlab, C, or SystemC. The basis of our techniques is the generation of the timestamp data for the frequency reference (FREF) and the DCO variable output

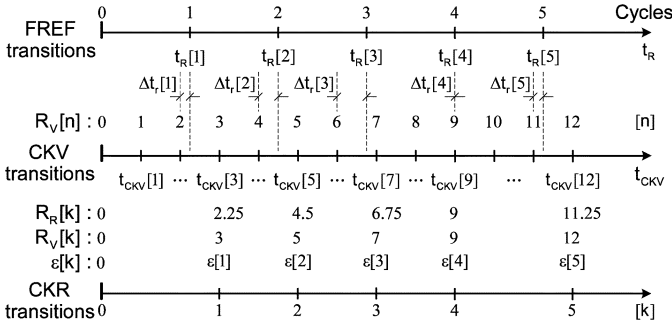


Fig. 6. ADPLL simulation principles: timing diagram. The development of the phase signals $R_R[k]$, $R_V[k]$ and $\varepsilon[k]$ pertaining to the ADPLL phase detection mechanism, is shown for $FCW = 2.25$.

(CKV) clock domains. Two progressively increasing integer-valued time indexes, k and n , are utilized to indicate the completion of full FREF and CKV cycles, respectively (see Fig. 6). Starting with index k , we generate the actual FREF clock transition as: $t_R[k] = k \cdot T_R + TDEV_{PN_R}[k]$, where T_R is the ideal FREF clock period and $TDEV_{PN_R}[k]$ is the FREF transition perturbation due to the reference clock phase noise. Given $t_R[k]$, we generate the actual CKV clock transitions up to $t_R[k]$, as $t_{CKV}[n] = n \cdot T_0 - TDEV[n] + TDEV_{PN}[n]$. At this point, the timing difference, $\Delta t_r[k]$ between the current rising FREF and CKV transitions (events) is readily available for the estimation of the fractional phase error ($\varepsilon[k]$) that needs to be compensated due to the retiming process of the reference (FREF) clock (see Fig. 2). The latter is estimated in hardware as follows. A TDC composed of a chain of inverters is employed for the measurement of the timing difference Δt_r in integer multiples of single inverter delays, Δt_{res} (in seconds) [see Fig. 7(a)] [4]. The number of TDC taps is determined by the number of inverter delays that are required to cover a full CKV clock period under the strong process corner (e.g., 15 ps for a 90-nm CMOS process). Due to the limited resolution of the TDC, only a quantized value of the actual timing difference Δt_{r_Q} is obtained. Subsequently, it is normalized to the CKV clock period, and the fractional phase error is given as

$$\varepsilon[k] = 1 - \frac{\Delta t_{r_Q}[k]}{T_V} \quad (3)$$

where $\bar{T}_V$ is obtained after long-term averaging of the hardware-based estimation of the DCO output (CKV) clock period T_{V_HW} as $\bar{T}_V = 1/N_{avg} \cdot \sum_{l=1}^{N_{avg}} T_{V_HW}[l]$.

To model the above TDC behavior, the behavioral model shown in Fig. 7(b) is employed, based on which the timing difference Δt_r is quantized with the resolution of Δt_{res} . The fractional phase error is then given as

$$\varepsilon[k] = 1 - \frac{1}{\bar{T}_V} \cdot \Delta t_{res} \cdot \left\lfloor \frac{\Delta t_r[k]}{\Delta t_{res}} \right\rfloor \quad (4)$$

where $\bar{T}_V$ is obtained after averaging of the single-shot CKV clock period as $\bar{T}_V = 1/N_{avg} \cdot \sum_{l=1}^{N_{avg}} (t_{CKV}[l] - t_{CKV}[l-1])$.

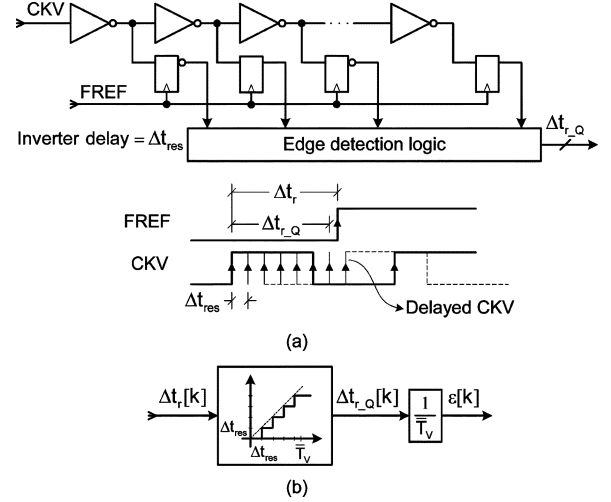


Fig. 7. TDC. (a) Simplified hardware implementation. (b) Behavioral model.

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FOR each FREF cycle
  Accumulate FCW:  $R_R[k] = R_R[k-1] + FCW$ 
  Generate FREF clock transition:  $t_R[k] = k \cdot T_R + TDEV_{PN\_R}[k]$ 
  WHILE  $t_{CKV}[n] < t_R[k]$ 
    OTW separation
     $OTW[k] = \text{FLOOR}(OTW[k-1])$  (Integer part)
     $OTW_F[k] = OTW[k-1] - OTW[k]$  (Fractional part)
     $\Sigma\Delta$  Modulator
    Apply  $OTW_F$  to the  $\Sigma\Delta$  modulator (Dithering of the DCO varactors)
    DCO
    CKV period deviation due to  $OTW[k]$ :  $TDEV_I[n]$ 
    CKV period deviation due to  $OTW_F[k]$ :  $TDEV_F[n]$ 
    CKV period deviation due to DCO phase noise:  $TDEV_{PN}[n]$ 
    Generate CKV clock transition:
     $t_{CKV}[n] = n \cdot T_0 - (TDEV_I[n] + TDEV_F[n]) + TDEV_{PN}[n]$ 
    Accumulate a CKV cycle:  $R_V[n] = R_V[n-1] + 1$ 
  END
  Calculate the timing difference  $\Delta t_{r\_Q}[k]$  between  $t_R[k]$  and  $t_{CKV}[n]$ 
  Calculate the fractional phase error:  $\varepsilon[k] = 1 - \Delta t_{r\_Q}[k] / T_{V\_avg}$ 
  Phase Detector
   $\phi_E[k] = R_R[k] - (R_V[k] - \varepsilon[k])$ 
  Loop Filter
  Apply  $\phi_E[k]$  to the loop filter to generate the normalized tuning word:  $NTW[k]$ 

  DCO gain estimation:  $K_{DCO\_est}$ 
  Generate the current oscillator tuning word:  $OTW[k] = \text{FREF} / K_{DCO\_est} \cdot NTW[k]$ 
END

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Fig. 8. ADPLL modeling principles: pseudocode.

On the basis of the timing diagrams shown in Fig. 6 and the aforementioned time-domain modeling techniques, we developed an algorithm that accurately models the ADPLL behavior. A pseudocode for this algorithm is shown in Fig. 8. The validity of the proposed modeling techniques is proven through the close agreement between the simulation and measurement results.

VI. SIMULATION RESULTS AND LAB VALIDATION

In this work, we used Matlab to model and simulate the phase-domain ADPLL-based frequency synthesizer with a proportional-plus-integral ($P + I$) loop filter only. This type-II second-order loop configuration provides 20 dB/dec low-pass and 40 dB/dec high-pass filtering of the TDC and DCO phase noise, respectively. The ADPLL order may be increased by incorporating a cascade of first-order low-pass infinite-impulse response (IIR) filters in series with the ($P + I$) filter [2].

The simulation performance using a Pentium 4 PC at 3.2 GHz is approximately 50 reference clock (FREF) cycles ($\sim 2 \mu s$) per

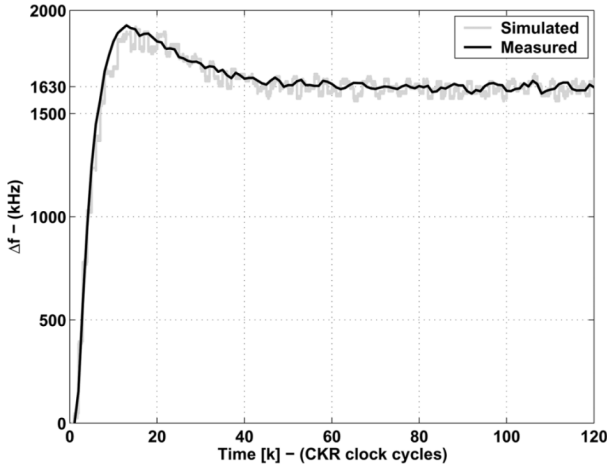


Fig. 9. Measured (using a Rohde and Schwarz FSIQ-7 signal analyzer connected to the RF port) and simulated trajectories of the DCO instantaneous frequency deviation for a frequency step of 1.63 MHz away from 1880 MHz. Loop parameters are per Table I.

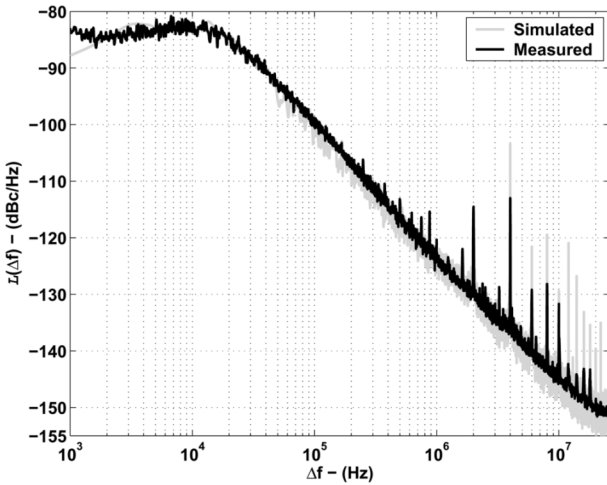


Fig. 10. Measured (using an HP8563E spectrum analyzer) and simulated (post-processing of the CKV timestamps: fast Fourier transform with a 2^{22} -points Blackman window) ADPLL output phase-noise spectrum. Loop parameters are per Table I. In this measurement, the IIR filters are turned off to show that spurs are also well predicted.

0.5 s of CPU time, which is many orders of magnitude faster than that of SPICE or SpectreRF [6]. A further decrease in simulation time can be achieved when precompiled programming languages are used, e.g., C or SystemC.

Fig. 9 shows the simulated and measured trajectories of the instantaneous frequency deviation of the ADPLL when only the tracking bank varactors are engaged. High-speed $\Sigma\Delta$ dithering of the varactors is enabled, and the DCO gain (K_{DCO}) was assumed to be constant and equal to 12 kHz/LSB. During this frequency settling, a larger loop bandwidth is used. Later, during the actual use, the ADPLL is hitlessly gear-shifted to a lower bandwidth in order to improve the noise performance. The simulated output phase-noise spectrum is shown in Fig. 10 superimposed on the measured phase-noise spectrum of the actual commercial ADPLL-based frequency synthesizer, proving the validity and effectiveness of the proposed modeling techniques. Table I lists the ADPLL simulation parameters.

TABLE I
ADPLL SIMULATION PARAMETERS

TYPE-II SECOND-ORDER ADPLL CONFIGURATION	
Reference frequency - f_R	26 MHz
FCW	72.307692
DCO output frequency - f_V	1880 MHz
Proportional gain α	2^{-2} (Fig. 9), 2^{-7} (Fig. 10)
Integral gain ρ	2^{-6} (Fig. 9), 2^{-16} (Fig. 10)
Loop bandwidth - f_{BW}	40.25 kHz (Fig. 10)
TDC resolution - Δt_{res}	20 ps
$\Sigma\Delta$ modulator	MASH 1-1 clocked at $\text{CKVD} = \frac{f_V}{8}$
DCO PHASE NOISE PROFILE	
Upconverted $1/f$ noise	-110 dBc/Hz at $\Delta f_F = 200$ kHz
Upconverted thermal noise	-145 dBc/Hz at $\Delta f_W = 10$ MHz
Measured thermal noise floor	-162 dBc/Hz

VII. CONCLUSION

We have proposed new event-driven time-domain modeling and simulation techniques and applied them to a phase-domain ADPLL-based RF frequency synthesizer. Generated models based on these techniques accurately predict the ADPLL transient, steady-state, and phase-noise performance by exercising the time-edges of interest only (the rising edges of the reference and DCO variable output signals). This results in a very short simulation time and limited amount of data for system performance evaluation. The accuracy of the proposed method is demonstrated by close agreement with the measured results.

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